

Matthew Hale

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EDUCATION

East Carolina University

Greenville, NC

Bachelor of Science in Biochemistry (Chemistry Conc.), Minor in Mathematics

Aug. 2024 – May 2026

GPA: 3.94 / Summa Cum Laude — Outstanding Senior Award — Chancellor's List Fall 2024, Fall 2025, Spring 2026;

Dean's List Spring 2025 — Moldin Scholarship Recipient — Quantum Mechanics Lecture Award

Wake Technical Community College

Raleigh, NC

Associate of Science

Aug. 2022 – May 2024

EXPERIENCE

NMR Research Assistant

Jan. 2026 – May. 2026

East Carolina University

Greenville, NC

- Collaborated with PhD researchers across multiple labs to develop NMR data processing workflows for PFAS-protein binding studies.
- Automated T1 inversion-recovery and chemical shift perturbation analyses, standardizing integration in calculations to improve reproducibility of T1 and binding constant (Kd) measurements.
- Presented findings to PI and graduate researchers, demonstrating improved consistency and efficiency in NMR data analysis.

Teaching Assistant

Aug. 2025 – Dec. 2025

East Carolina University

Greenville, NC

- Assisted in running introductory biology laboratory sections, guiding 40+ students through experimental procedures, data collection, and data analysis using Excel.

Bioinformatics Researcher

May 2025 – Sept. 2025

East Carolina University

Greenville, NC

- Expressed and characterized CsgA protein in E. coli through transformation, bacterial culture, cell lysis, and SDS-PAGE, confirming expression via His-tag fluorescence staining.
- Cleaned and analyzed mass spectrometry proteomics data in Excel and SQL using filtering and statistical transformations to identify protein expression changes.
- Developed a Python-based bioinformatics pipeline to automate large-scale protein analysis across UniProt, ProtParam, BLAST and PDB databases.
- Restored a Shimadzu RF-5301 fluorometer to operation through motherboard replacement and firmware recovery via PuTTY.

PROJECTS

NMR Analysis Suite | *PyQt6, NMRglue, SciPy, NumPy, Matplotlib*

Mar. 2026 - May. 2026

- Developed a Python application to automate T1 relaxation and chemical shift perturbation analysis for PFAS-protein binding studies, replacing manual NMR calculations performed by lab researchers.

ProtMerge | *Tkinter, Pandas*

May 2025 - Sept. 2025

- Built a bioinformatics pipeline integrating UniProt, BLAST, and PDB to automate large-scale protein characterization across 8,000+ human and bacterial proteins.

Polymerization Kinetics | *UV-Vis, Fluorescence Spectroscopy*

Spring 2026

- Designed a laboratory experiment investigating ethyl cyanoacrylate polymerization kinetics via UV-Vis and fluorescence spectroscopy, extracting pseudo-first-order rate constants and connecting kinetic behavior to polymer structure and polydispersity.

TECHNICAL SKILLS & COURSEWORK

Familiar Laboratory Instrumentation: HPLC, GC-MS, UV-Vis, Fluorescence Spectroscopy, Bomb Calorimetry, Polarimetry, Enzymatic Assays, SDS-PAGE, TLC

Programming & Data Analysis: Python, SQL, Excel

Relevant Software Experience: PyMOL, Gaussian, ORCA, Autodock Vina, Linux

Mathematical & Scientific Foundation: Differential Equations, Statistics, Quantum Mechanics, Thermodynamics, Quantitative and Instrumental Analysis, Biochemistry I-II, Organic Chemistry I-II, Genetics